

Biology Practice Test 04 (Evening Session)

Format: High-Complexity Board Simulator

Portion: Set 1 Midterm (Remaining Untouched Syllabus)

Time Allowed: 2 Hours (*Speed & Stamina Test!*)

Maximum Marks: 80

General Instructions:

- **CRITICAL RULE:** Answer ALL parts of a question. Do not ignore the second sentence of a prompt.
 - **NO BLANKS ALLOWED:** A -1 mock penalty applies for leaving any question blank. Use the Brain Dump technique if unsure.
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SECTION A (15 Marks)

Answer all questions. Each question carries 1 mark.

Question 1: Name the specific diagnostic blood test used to confirm Typhoid fever.

Question 2: Name the ecological rule which states that mammals from colder climates generally have shorter ears and limbs to minimize heat loss.

Question 3: Which genus of viruses is widely used as excellent biological control agents against insects and other arthropods?

Question 4: Name the chromosomal disorder caused by the presence of an additional copy of the X-chromosome resulting in a karyotype of 47 (44A + XXY).

Question 5: What is the specific genetic term for a cross between an F_1 hybrid and its homozygous DOMINANT parent?

Question 6: Name the pituitary hormone that causes strong uterine contractions during parturition.

Question 7: Name the non-cellular layer that surrounds the mammalian ovum immediately outside the plasma membrane.

Question 8: What is the surgical sterilization method used in human females called?

Question 9: Which specific human blood group is considered the Universal Acceptor?

Question 10: Name the fungus from which the immunosuppressive drug Cyclosporin-A is extracted.

Question 11: Give one example of an ecological interaction representing Commensalism.

Question 12: What is the process of formation of megaspores from the megaspore mother cell (MMC) called?

Question 13: The evolutionary process giving rise to new species adapted to new habitats and ways of life, starting from a common ancestor (e.g., Darwin's finches), is called what?

Question 14: What is the phenomenon called when two alleles of a gene are completely and equally expressed in a heterozygous individual (e.g., AB blood group)?

Question 15: Name the specific species of *Plasmodium* that causes the most severe and fatal form of malignant malaria.

SECTION B (20 Marks)

Answer all questions. Each question carries 2 marks.

Question 16: Differentiate between Commensalism and Amensalism.

Question 17: What are Statins? Name the microbe they are commercially extracted from.

Question 18: Differentiate between Vasectomy and Tubectomy.

Question 19: Explain the exact molecular cause (point mutation) of Sickle Cell Anemia, specifying the change in the amino acid sequence of the beta-globin chain.

Question 20: Why do people visiting high altitudes (like Rohtang Pass) initially experience altitude sickness, and what are two ways the human body naturally acclimatizes to it?

(Do not skip the second part!)

Question 21: State the ploidy (n, 2n, 3n) of the following structures in an angiosperm: (i) Nucellus, (ii) Microspore tetrad, (iii) Endosperm, (iv) Synergids.

Question 22: Differentiate between Incomplete Dominance and Co-dominance.

Question 23: State the specific functions of: (a) The Acrosome of a sperm, (b) The Leydig cells of the testis.

Question 24: What are biofertilizers? Give one example of a symbiotic nitrogen-fixing bacterium.

Question 25: Define Parthenocarpy. Give one example of a naturally parthenocarpic fruit.

SECTION C (30 Marks)

Answer all questions. Each question carries 3 marks.

Question 26: Draw a neat, well-labeled diagram of an Antibody molecule (IgG).

Question 27: Briefly describe the replication cycle of a retrovirus (like HIV) inside a human macrophage.

Question 28: Explain the chromosomal mechanism of sex determination in birds. How does it differ fundamentally from sex determination in humans?

(Do not skip the second part!)

Question 29: A woman with blood group O marries a man with blood group AB. Show the possible blood groups of their children using a Punnett square. What is the probability of a child having blood group O?

Question 30: Differentiate between Microsporogenesis and Megasporogenesis (provide three points of difference).

Question 31: Describe the structure of a mature, 7-celled, 8-nucleate embryo sac (female gametophyte) of an angiosperm.

Question 32: Explain the mechanism of action of oral contraceptive pills (how do they prevent pregnancy?).

Question 33: Write short notes outlining the chromosomal cause and two physical symptoms for: (a) Down's Syndrome, (b) Turner's Syndrome.

Question 34: Discuss the role of Mycorrhiza (fungal association with plant roots) as a powerful biofertilizer.

Question 35: Differentiate between homologous and analogous structures by providing **plant** examples for each.

SECTION D (15 Marks)

Answer any ONE question from each of the three pairs below.

Question 36: (Option A) Trace the complete life cycle of *Plasmodium* (Malarial parasite), detailing the stages that occur in the human host and the stages that occur in the mosquito host. (5 Marks) **OR (Option B)** Describe the various stages of embryonic development in humans, starting from fertilization, cleavage, morula formation, up to the formation of the blastocyst and its implantation in the uterus. (5 Marks)

Question 37: (Option A) Explain the inheritance pattern of Hemophilia (a sex-linked recessive disease). Use a cross between a carrier woman ($X^H X^h$) and a normal man ($X^H Y$). Why is this disease extremely rare in human females? (5 Marks) **OR (Option B)** Describe the specific physiological, behavioral, and structural adaptations of desert plants (xerophytes) and desert animals (like the Kangaroo rat) that allow them to survive extreme aridity. (5 Marks)

Question 38: (Option A) What is Cancer? Differentiate between benign and malignant tumors. Explain three major approaches used in the clinical treatment of cancer. (5 Marks) **OR (Option B)** Explain the complete process of Spermatogenesis in humans. Support your answer with a schematic representation showing the stages from spermatogonia to spermatozoa, including the ploidy at each stage. (5 Marks)